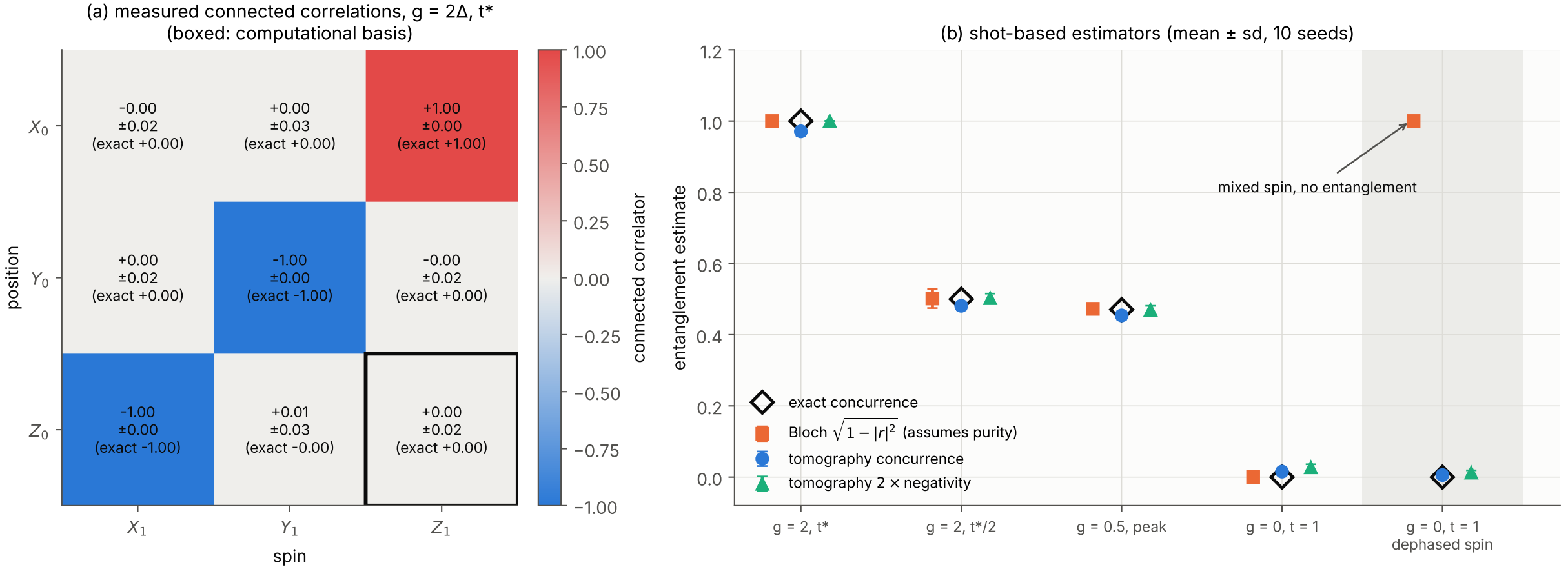
(a) measured connected correlations, g = 2Δ, t*
(boxed: computational basis)
-0.00
±0.02
(exact +0.00)
+0.00
±0.03
(exact +0.00)
+1.00
±0.00
(exact +1.00)
+0.00
±0.02
(exact +0.00)
-1.00
±0.00
(exact -1.00)
-0.00
±0.02
(exact +0.00)
-1.00
±0.00
(exact -1.00)
+0.01
±0.03
(exact -0.00)
+0.00
±0.02
(exact +0.00)
X_0
Y_0
Z_0
position
X_1
Y_1
Z_1
spin
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
connected correlator
(b) shot-based estimators (mean ± sd, 10 seeds)
entanglement estimate
1.2
1.0
0.8
0.6
0.4
0.2
0.0
mixed spin, no entanglement
exact concurrence
Bloch $\sqrt{1-|r|^2}$ (assumes purity)
tomography concurrence
tomography 2 × negativity
g = 2, t*
g = 2, t*/2
g = 0.5, peak
g = 0, t = 1
g = 0, t = 1
dephased spin